Analysis and the model

Index → features → target → baseline → limits

What you need to know

- The Champagne Index formula: recorded goals plus assists, divided by the matches the source reports for that club-season. Because goals and assists are counted separately, one goal with an assist contributes twice. It is our own proxy for attacking output, not xG and not an official rating.
- The target is the next consecutive season's index minus the current one. A season pair only counts with at least 30 matches in both seasons for the top five leagues, and five for other competitions.
- The five features: the current index, total incoming spend, spend on attacking players, average squad age, and the change in average market value.
- How the wrong lag was removed: build one row per club and season first, attach the exact previous year, and only then join competitions. Done in the other order, a club playing three competitions is counted three times before the lag is computed.
- The chronological holdout: earlier seasons train the model, later seasons test it, and the baseline predicts no change at all on those same test rows. A random split would let future seasons leak into training.
- $R^2 = 0.127$ does not mean 12.7% accuracy, and it is not a share of correct predictions. The honest comparison is MAE 0.4856 against the baseline's 0.5114 — about five percent better.
- The Lab uses a separate model fitted on all eligible data for exploration. Its own fit is not an independent test, and moving a budget slider shows a learned association, not the effect of an actual transfer.
- The season comparison also shows goals for, goals against, goal difference and per-match attack and defense ranks derived from real scorelines. Those are separate from the Transfermarkt-based index.
- The defensive-spend follow-up correlates defender fees with the next-season change in goals conceded per match: r = 0.0040, p = 0.427, n = 38,531. There is no useful linear signal.

What to show

1. Open the Project page and read the current r, p, n and MAE aloud, naming the units and the sample behind each one.
2. Explain why a 23/24 → 25/26 pair is invalid even though a naive shift by one row returns it.
3. Change one input in the Lab, return to the baseline, and say what the comparison between the two actually measures.

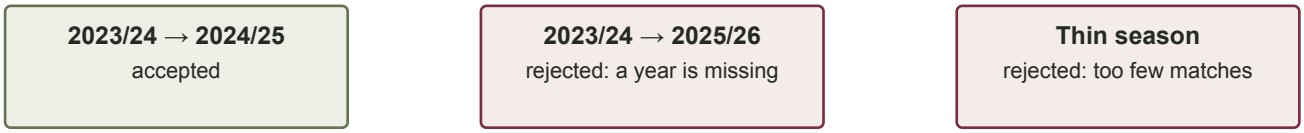
Your part in pictures

What the model is asked to predict

How to read it. The prediction target is a change between two seasons that genuinely follow each other, not a league position and not a match result.



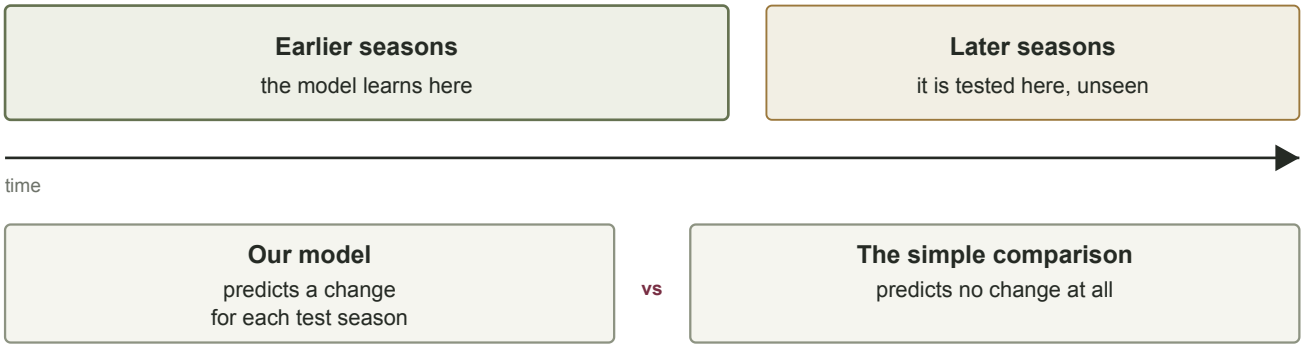
A pair is only used when the two seasons really follow each other:



> **Say it like this:** “We predict how a club's attacking output changes in the following season — and only when both seasons are properly covered.”

Learning on the past, tested on the future

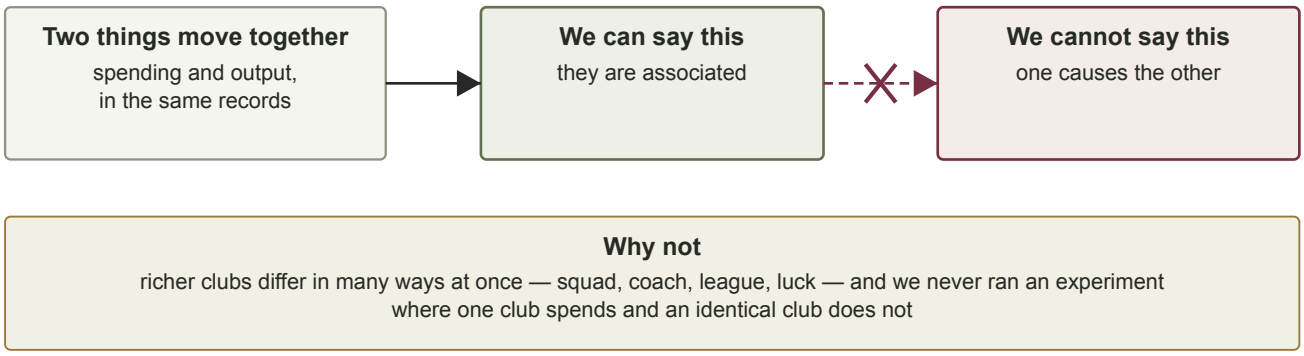
How to read it. Time runs left to right. The model never sees the seasons it is judged on, and it is compared with the simplest possible guess.



> **Say it like this:** “We train on earlier seasons and test on later ones, and we compare the model with simply predicting no change at all.”

What we may and may not claim

How to read it. The left arrow is allowed, the crossed one is not. This is the boundary every number in our project sits behind.



> **Say it like this:** “These are historical records, not an experiment, so we can say two things move together — never that one causes the other.”

Your spoken part

> *We define attacking output as recorded goals plus assists per source match. The target is the change in that index in the next consecutive season. We remove incomplete season pairs and construct lagged squad values before joining competitions. Evaluation uses earlier seasons for training and later seasons for testing. The RandomForest has a test MAE of about 0.486, compared with 0.511 for predicting no change: an improvement of roughly five percent. This is a modest signal. The comparison also separates the Transfermarkt attacking index from completed league scores: goals for, goals against, their difference, and per-match attack and defense ranks. A defensive-spend follow-up also has almost zero association with next-season conceded-goal change (r=0.004, p=0.427; n=38,531). The Lab uses a separate all-data model for retrospective exploration. Moving a budget slider changes a model input; it does not establish what a transfer would cause.*

Hand over: “The application makes those distinctions visible and lets us inspect the evidence for individual clubs.”

Your slots on the clock

Slide	Block	Time	Length
4. Define the observation before fitting	04 / Method	03:45–05:15	90 s
5. An honest model comparison	05 / Evaluation	05:15–06:30	75 s
7. What the data supports	07 / Findings	09:15–10:15	60 s

How to close the exam

1. You own slides 4 and 5 (03:45–06:30) and slide 7 (09:15–10:15). Jacopo's 165-second demo sits between them, so watch the clock or you will lose your findings slide.
2. Define the observation before showing the model: club × season × competition, then the target as the difference against the next consecutive season.
3. Say the headline number as a comparison, not an achievement: MAE 0.4856 against a baseline of 0.5114, roughly five percent better.
4. Explain immediately that $R^2 = 0.127$ is not a share of correct predictions. It is the most likely question on your part.
5. If asked about the Lab: it is a separate all-data model for exploration, and its own fit is not an independent test.
6. Use “accompanies” and “is associated with” for every relationship. The word “causes” is more expensive on your part than anywhere else.
7. On xG, answer briefly: the provider field does not exist, we show real shots, and the shots-on-target estimate is an experimental proxy.

Not your part — hand it over

1. Infrastructure numbers are not yours: startup time, memory, the index speed-up and the growth experiment are Atai's part. Hand them over in one sentence.
2. Data provenance, acquisition and the schema are Neda's part. You start from what is already in the tables.
3. If asked for a number that is not in this guide, name the file rather than a remembered value: docs/findings.md and docs/LAB.md.

Where to look in the project

Project file	What to look at in it
`pipeline/compute_predictions.py`	The target, the five features, the market-value lag, the chronological holdout and the zero-change baseline — all in one file.
`docs/LAB.md`	The model protocol and the difference between the Lab model (all data) and the evaluation model (earlier seasons → later seasons).
`docs/findings.md`	The current r, p, n, R^2 and MAE. Read the numbers from here rather than from memory.
`docs/GLOSSARY.md`	Definitions of the Champagne Index, the Discipline Index, the ranks, and the xG boundary.
`pipeline/feature_correlations.py`	The feature correlation matrix and the collinearity flag above the threshold.

Project file	What to look at in it
`pipeline/shap_analysis.py`	Mean |SHAP| computed on the same held-out split as the R^2, which answers “why is this not in-sample?”
`pipeline/power_ratings.py`	Poisson attack and defense ratings with a shared home advantage; the code states that the estimate is in-sample.
`pipeline/proxy_xg.py`	The shots-on-target proxy with finishing and defensive luck, and why none of it is xG.
`pipeline/discipline_effect.py`	The red-card association with conceded goals: +0.341 per team-match, 1,619 observations against 16,195.
`tests/test_regressions.py`	Known defects in the model and the calculations, pinned so they cannot come back.

Plain-language glossary

These definitions are deliberately non-technical. Use them to explain the system to someone outside the course, and to answer a question whose wording you do not recognise.

Container / Docker Compose	A container is one packaged program with everything it needs to run. Compose is the file that starts our four containers in the right order: the database first, then the loader, then the processing pipeline, then the web application.
Relational database / PostgreSQL	A store of tables with fixed columns that can be combined by shared keys. We use PostgreSQL 16. Combining two tables on a shared key is called a join.
Schema	The declared shape of the tables: which columns exist and what type each one holds. It is written in sql/schema.sql.
COPY and index	COPY loads a whole file into a table in one stream. An index is a lookup structure, like the index of a book: it makes selective queries fast and is built after the bulk load.
Pipeline	The processing step between the database and the web page. It reads tables, computes our measures and the model, and writes prepared result tables back.
API / endpoint / JSON	The API is the set of web addresses the page calls. Each address is an endpoint. JSON is the plain-text format of the answer.
Champagne Index	Our own measure of attacking output: recorded goals plus assists, divided by the number of matches the source reports for that club-season. It is a proxy, not an official statistic.
Target and feature	The target is what the model tries to predict — here the change in the index in the next consecutive season. A feature is an input the model uses to predict it.
Chronological holdout	We train the model on earlier seasons and test it on later seasons it has never seen. A random split would mix the future into the training data and flatter the result.

Baseline	The simplest possible prediction, used for comparison. Ours predicts no change at all. A model is only useful if it beats the baseline on the same test rows.
R^2 and MAE	R^2 is the share of variation the model explains, from 0 to 1 — ours is 0.127, which is weak. MAE is the average absolute error in index units — ours is 0.4856 against a baseline of 0.5114, about 5% better. Neither is a percentage of correct answers.
Correlation vs. causation	A correlation (r) measures how two numbers move together, from −1 to +1. It cannot show that one causes the other. Every result in this project is a correlation, including the red-card and spending findings.
Proxy	A measurable stand-in for something we cannot measure directly. Our shots-on-target estimate is a proxy for chance quality; it is not xG, which the data source does not contain.
Poisson rating	A simple statistical model of how many goals each team scores and concedes, fitted to one league-season. It describes that season; it does not forecast new matches.
SHAP	A method that attributes a model's prediction to its individual inputs, so we can report which feature mattered most without relying on the model's internal importance score.

Practice

1. Compute (76 + 70) / 38 and explain why 3.84 is not goals per match.
2. Compute (0.511364 − 0.485593) / 0.511364 ≈ 5.04% and state the correct interpretation out loud.
3. Separate three questions: a correlation inside one season, a change in the next season, and the causal effect of buying a player.

Questions for your part

Is Champagne Index an official metric or xG?

No. It is this project's proxy: recorded goals plus assists divided by source matches. A goal with an assist contributes twice. We explain the denominator and do not present it as xG or an official rating.

What exactly does the model predict?

The next consecutive season's Champagne Index minus the current index. It does not predict a match score, a calibrated league position or the causal return from buying a player.

Why reject gaps between seasons?

A club may leave a competition and return years later. shift(-1) gives the next recorded row, which is not necessarily next year. We explicitly require a one-year gap and adequate coverage in both seasons.

Why use a temporal split?

A random row split mixes earlier and later seasons and can make historical evaluation overly optimistic. We train on earlier seasons and test on later ones, keeping observations from the same season together.

How much better is the model than a baseline?

On the same 5,585 held-out rows, MAE is about 0.4856 versus 0.5114 for predicting no change. That is about a five percent reduction in MAE, a modest gain. R-squared is about 0.127, not a percentage of correct predictions.

Can you conclude that buying attackers improves performance?

No. These are retrospective observational data with confounding and source limitations. A slider changes a model input and shows a learned association, not the outcome of a controlled transfer intervention.

Why is the Lab model different from the evaluation model?

The evaluation model uses only earlier training data and is scored on held-out later seasons. A separate model is refitted on all eligible data for exploration. Its own training fit is not claimed as independent test performance.

Why did the index-position correlation change after the audit?

The earlier result mixed a partial 25/26 attacking snapshot with full final standings. We now require adequate comparable coverage. The corrected same-season association is about minus 0.793 over 388 observations.

What is the main result in one sentence?

The system is reproducible on the measured setup, while historical transfer spending alone has almost no linear relationship with next-season index change and the richer model adds only modest predictive value.

What is the most important limitation?

The data are retrospective and unevenly covered, the index is a proxy, and associations are not causal effects. Source licence and independent team/laptop validation also remain explicit unresolved requirements.

How do goals for, goals against and goal difference work in the comparison?

GF and GA are season totals rebuilt from completed home/away match scores. GD is GF − GA. They come from the optional football-data.co.uk match enrichment, not from the Champagne Index player totals.

How are attacking and defensive ranks calculated?

Within one league and season, attack rank orders goals scored per match from highest to lowest; defense rank orders goals conceded per match from lowest to highest. SQL RANK gives tied rates the same rank and leaves a gap after ties. They describe scores, not tactics or chance quality.

Do we have xG or can we estimate it?

The source has no provider xG field. The comparison instead shows observed shots and shots on target. A separate estimate multiplies shots on target by the league-season average goals-per-shot-on-target rate; it is an experimental volume proxy with no shot-location or chance-quality input, not xG.

What is the difference between the Champagne Index and attack rank?

The Champagne Index is recorded player goals plus assists divided by a Transfermarkt matches proxy for a club-season. Attack rank uses actual completed league scorelines, sums team goals and ranks goals per match within the league-season. Their sources, coverage and units differ.

How did you check the goalkeeper defensive totals?

We compared Transfermarkt goalkeeper goals-conceded totals with the independent match-result totals on 388 adequately covered top-five club-seasons. They match exactly in 366 (94.3%). Clean sheets match only 269 (69.3%), so the scoreline source remains the basis of league ranks.

What does the defensive-spend analysis show?

The Pearson correlation between defender transfer fees and next-season change in the goalkeeper-source conceded-goal rate is r=0.004047, p=0.426997, n=38,531. The sample does not support a useful linear association. It is descriptive, not causal. A separate match-level check finds teams with a red card concede 0.341 more goals on average in the same match, but match context confounds that association.

Why keep the player-source defensive values separate from scoreboard goals?

The sources use different records and denominators. Goalkeeper conceded totals agree with scorelines in 94.3% of adequately covered seasons, while clean sheets agree in 69.3%. The UI therefore uses match scores for GF, GA, GD and ranks; player-source totals remain a quality cross-check. Shots and corners are also observed match statistics, not chance-quality measures.

Is the shot-based estimate the same as xG?

No. It multiplies a team's shots on target by the league-season average goals per shot on target. It has no shot-location, distance, body-part, pressure or goalkeeper-position input. It is a rough volume benchmark and must be called an SOT-based proxy, not provider xG.

What does the Poisson attack/defense rank tell us?

The model fits attack, defense and a shared home-advantage term to the selected league-season scores. Larger fitted attack and defense parameters mean stronger estimated attack and defense. Its ranks are descriptive and in-sample; the separate chronological RandomForest evaluation is the project's held-out prediction check.

Did the added defensive features improve the model?

On the same chronological split, R^2 moves from 0.1271 with the five-feature baseline to 0.1502 after adding goalkeeper conceded rate, discipline and net spend; MAE moves from 0.4856 to 0.4788 on 5,585 held-out rows. The gain is modest, and it does not isolate a causal effect of defense.

Do red cards cause more goals to be conceded?

The local sample shows a same-match difference of +0.341 conceded goals per team-match (1,619 sent-off vs 16,195 others; Welch $p < 0.0001$). This is observational: stronger opponents, match state and timing can affect both red cards and goals. It is not a causal estimate.

Phrasing to avoid

- The model proves that a bigger budget improves performance.
- R^2 is the share of correct predictions.
- The held-out evaluation is a test of the all-data Lab model.

Exam day

Date	Format	What is required
12 November 2026 CIDS Lecture 15, pp. 37–39	Checkpoint · 5 minutes	One end-to-end demo, one visible result and the first measured full-run time. A polished interface is not required.
20 / 26 November 2026 CIDS Lecture 15, pp. 37–39	Final defense · 15 minutes + 5 minutes of questions	Every real participant speaks. Suggested split: question and data 3 min, architecture 2, live demo 3, measurements 3, problems 2, contributions 2.

Machine and materials

1. Thirty minutes ahead: run `docker compose up -d --build`, then open /api/health and wait for `ready=true` with the loader and pipeline finished.
2. Open four tabs in advance: Overview (Serie A 24/25), the Inter / Milan comparison, the Lab, and the Project page with the numbers.
3. Open the offline study site `defense/site/index.html` and `defense/site/downloads/Presentation.pdf`: both work without a network and without the database.
4. Check that the screenshots in `defense/site/assets/*-desktop.png` are present. They are the fallback if Docker does not come up.
5. Turn off notifications, connect the charger, test screen mirroring and increase the browser font size.
6. Decide in advance who drives the demo and who watches the clock. The timer is built into `defense/site/presentation.html` (N for notes, F for fullscreen).

How to answer questions

- Answer in three steps: one short sentence, then the evidence (a file or a number), then the limitation.
- Give every number with its sample and unit: “r = −0.79 across 388 club-seasons, within the same season”, not “the correlation is high”.
- If you do not know, say where the answer lives instead of guessing: “that is in docs/MEASUREMENTS.md, the memory sampling method is described there”.
- If the question belongs to someone else's part, hand it over in one sentence rather than approximating their area.
- Never turn an association into a cause. Say “accompanies” or “is associated with”, never “causes” or “improves”.
- If asked about a weak point, answer it first and directly. An acknowledged limitation is stronger than a defended inaccuracy.

Uncomfortable questions: answer them first

Topic	Question	Answer and evidence
Licence of the primary source	Is your data openly licensed?	We pinned the source revision and verified headers and checksums, but there is no licence file, so we did not establish an explicit reuse licence. We document it as unresolved and we do not redistribute the raw data. Public downloadability is not a licence. `docs/SOURCES.md, docs/LICENCE_DECISION.md`
Four contributors	Who did what in this project?	Describe only the work you can open and explain on screen. Git identities and role cards are not proof of independent contribution, and the repository says so. `docs/WORK_PACKAGES.md, README.md`
Reproduction on another machine	Did someone else reproduce it?	The run is reproducible on the measured machine, and we also rebuilt it from a clean clone. An independent fresh checkout on another person's laptop has not been recorded yet, so we do not claim it. `docs/FRESH_CHECKOUT.md, docs/MEASUREMENTS.md`
xG and proxies	Is this xG?	No. The source has no provider xG field. We show observed shots and shots on target, and a shots-on-target conversion proxy that we label experimental. The Poisson attack and defense ratings describe the season they are fitted on. `docs/GLOSSARY.md, pipeline/proxy_xg.py, pipeline/power_ratings.py`

> A role is a study responsibility. Do not claim authorship or a verification you did not perform. Show exactly the changes and commands you ran yourself.